

WestQuant Representation Scheduler (Project A)

Slide 1 — Project Overview

Problem: A fixed logical Hamiltonian does not determine a unique useful quantum representation. The physical embedding/layout matters.

Approach: Automated search over QoolQit embeddings (InteractionEmbedder, SpringLayoutEmbedder, Blade) using successive halving across 5 stages: cheap metrics, logical fidelity, compilation, emulation, robustness.

Results: Flagship experiment (6 problems, 9 embeddings, 3 replicates): embedding main effect $\eta^2(R) = 19.0\%$ (median). For 5/6 problems, the preselected baseline fails terminal ground-state preservation; search finds feasible alternatives for 6/6.

Importance: Treating the embedding as an optimization variable (not a fixed choice) materially improves quantum solution quality. Open-source, deterministic, no proprietary infrastructure required.

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Slide 2 — QoolQit Experience

- Positive aspects:** QoolQit's Register, Drive, and QuantumProgram abstractions are clean and composable. The embedding API (InteractionEmbedder, Blade) is well-designed. `compile_to(device)` handles device constraints automatically. LocalEmulator with QutipBackendV2 works out of the box.
- Challenges:** DMM drives require a DMM-capable device (AnalogDeviceWithDMM, not AnalogDevice). The `max_energy` compilation profile is needed for automatic rescaling; the default profile requires pre-scaled drives. Bitstring results come as Counters of strings (not arrays), requiring careful bit-ordering handling.
- Workflow:** The successive-halving approach (cheap metrics first, expensive emulation last) is essential for scaling the search. QoolQit's `interaction_matrix()` and $1/r^6$ convention made interaction-fidelity metrics straightforward.